Orbital Simulation Project

This project simulates satellite orbits and visualizes communication with the Iridium constellation. It features a Python Flask backend for orbital calculations and a React TypeScript frontend for 3D visualization.

Prerequisites

- Python 3.8 or newer
- Node.js 14 or newer
- Internet connection (for initial setup)

Quick Start

1. Run `setup.bat` to install dependencies and set up the project
2. Run `run.bat` to start the application
3. Open http://localhost:3000 in your browser

Project Structure

```
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├── backend/              # Python backend
│   ├── app.py            # Flask server
│   ├── orbital.py        # Orbital mechanics
│   ├── data/             # TLE data
│   └── venv/             # Python virtual environment
├── frontend/             # React frontend
│   ├── src/
│   │   ├── components/
│   │   │   ├── OrbitSimulator.tsx    # Main component
│   │   │   └── OrbitSimulator.css    # Component styling
│   │   ├── App.tsx      # React app
│   │   └── App.css      # App styling
│   └── ... (other React files)
├── setup.bat             # Setup script
└── run.bat               # Run script
```

Features

- Configure custom satellite orbit parameters

- Visualize orbits in 3D with interactive globe
- Select specific satellites to view
- Calculate communication windows and statistics
- Analyze connectivity percentages

Usage

1. **Parameters Tab**:
   - Select orbit type (sun-synchronous or non-polar)
   - Set altitude and inclination
   - Configure LTAN for sun-synchronous orbits
   - Click "Run Simulation" button
2. **Visualization Tab**:
   - Select satellites from the dropdown
   - View orbital paths on the 3D globe
   - See color-coded satellite markers
3. **Statistics Tab**:
   - Review orbital parameters
   - Check connectivity percentages
   - View communication windows

Development

- Backend code is in Python using Flask
- Frontend uses React with TypeScript
- 3D visualization uses react-globe.gl

Troubleshooting

- If the backend fails to start, check that port 5000 is available
- If the frontend fails to start, check that port 3000 is available
- If visualization is slow, try selecting fewer satellites

Credits

- Satellite TLE data from CelesTrak

- Earth imagery from NASA Visible Earth
- SGP4 orbital propagation algorithm